

Мощь средних

Схема матстата:

$$X_1, \dots, X_n \sim \text{iid } \mathcal{F}_\theta$$

Предпосылки
их надо всегда проверять.

Точечная оценка

$$\hat{\theta}$$

плотность

$$f_{\hat{\theta}}(t)$$

CI

AB-test

CI для $\hat{y}_{n+1}$
.....

→ метод моментов

$$\mathbb{E}(X) = f(\theta) \approx \bar{X}_n$$

$$\hat{\theta} = f^{-1}(\bar{X}_n)$$

→ ЦПТ

→ Дельта-метод

→ $t(n)$, χ^2 ,

→ Бутстрэп

.....

ОТВЕТЫ НА
ВОПРОСЫ

→ ММП

.....

Асимптотические г.ч.

$$X_1, \dots, X_n \sim \text{iid}(\mu, \sigma^2) \quad \sigma^2 < \infty$$

$$\hat{\mu}^{\text{MM}} = \bar{X}_n \quad \text{estimator - формула} \quad \text{estimate - число}$$

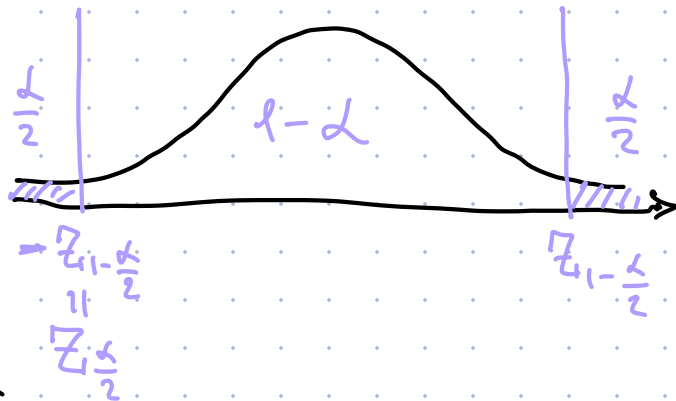
$$\bar{X}_n = \frac{X_1 + \dots + X_n}{n} \stackrel{\text{asy}}{\sim} \mathcal{N}\left(\mu; \frac{\sigma^2}{n}\right)$$

$$\mathbb{E}(\bar{X}_n) = \mathbb{E}\left(\frac{X_1 + \dots + X_n}{n}\right) = \frac{1}{n}(\mathbb{E}(X_1) + \dots + \mathbb{E}(X_n)) = \frac{1}{n} \cdot n \cdot \mu = \mu$$

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2}(\text{Var}(X_1) + \dots + \text{Var}(X_n)) = \frac{1}{n^2} \cdot n \cdot \sigma^2 = \frac{\sigma^2}{n}$$

Хочу понять методу какими $\hat{\mu}^L$ и $\hat{\mu}^R$ равен μ

$$Z = \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \stackrel{\text{asy}}{\sim} \mathcal{N}(0, 1)$$



$$P\left(-z_{1-\frac{\alpha}{2}} \leq \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \leq z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$P\left(\bar{X} - z_{1-\frac{\alpha}{2}} \cdot \frac{\hat{\sigma}}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{1-\frac{\alpha}{2}} \cdot \frac{\hat{\sigma}}{\sqrt{n}}\right) = 1 - \alpha$$

$\hat{\mu}^L$ $\uparrow$ не зная $\hat{\mu}^R$

Значо и чело
считать по
выборки

? Почему мы
можем без вреда
окопнать σ

уПТ:

$$\bar{X}_n \stackrel{\text{asy}}{\sim} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

СИ:

$$\bar{X}_n \stackrel{\text{asy}}{\sim} \mathcal{N}\left(\mu, \frac{\hat{\sigma}^2}{n}\right)$$

$$\underbrace{\frac{\bar{X}_n - \mu}{\frac{\hat{\sigma}}{\sqrt{n}}}}_{\mathcal{N}(0,1) \cdot 1} \cdot \frac{\hat{\sigma}/\sqrt{n}}{\hat{\sigma}/\sqrt{n}} = \underbrace{\frac{\bar{X}_n - \mu}{\hat{\sigma}/\sqrt{n}} \cdot \frac{\hat{\sigma}}{\hat{\sigma}}}_{\mathcal{N}(0,1) \cdot 1} \stackrel{P}{\rightarrow} \mathcal{N}(0, 1)$$

$$\boxed{\hat{\sigma} \xrightarrow{P} \sigma}$$

$$\hat{\sigma}^2 = \overline{X^2} - \bar{X}^2 \xrightarrow{P} E(X^2) - E(X)^2 = \sigma^2$$

$g(t) = \sqrt{t}$ шл. м.-в.-с.

$$\begin{array}{ccc} X_n & \xrightarrow{d} & X \\ Y_n & \xrightarrow{P} & a \end{array}$$

$$X_n \cdot Y_n \xrightarrow{d} a \cdot X$$

III. Slutsky

$$\begin{array}{ccc} Z_n & \xrightarrow{d} & N(0,1) \\ Y_n & \xrightarrow{P} & 1 \end{array}$$

$$\begin{array}{ccc} Z_n \cdot Y_n & \xrightarrow{d} & N(0,1) \\ \downarrow d & & \downarrow P \\ N(0,1) & & 1 \end{array}$$

Запам'ятати

$$X_1, \dots, X_n \sim \text{iid Bern}(p)$$

$$X_i \in \{0, 1\} \quad \hat{p} = \bar{X} = \frac{1}{n} \sum X_i$$

Большое n

$$E(\hat{p}) = \mu = p$$

$$\text{Var}(\hat{p}) = \frac{\sigma^2}{n} = \frac{p(1-p)}{n}$$

$$\hat{p} \overset{\text{asy}}{\sim} N\left(p, \frac{\hat{p}(1-\hat{p})}{n}\right) \quad P: \hat{p} \pm \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\begin{array}{l} X_1, \dots, X_n \sim \text{iid Bern}(p_1) \\ Y_1, \dots, Y_n \sim \text{iid Bern}(p_2) \end{array} \quad \left| \begin{array}{l} \text{незав. групу} \\ \text{от групу} \end{array} \right.$$

$$\hat{p}_1 \overset{\text{asy}}{\sim} N\left(p_1, \frac{\hat{p}_1(1-\hat{p}_1)}{n}\right) \quad \hat{p}_2 \overset{\text{asy}}{\sim} N\left(p_2, \frac{\hat{p}_2(1-\hat{p}_2)}{n}\right)$$

$$\hat{p}_2 - \hat{p}_1 \overset{\text{asy}}{\sim} N\left(p_2 - p_1, \frac{\hat{p}_1(1-\hat{p}_1) + \hat{p}_2(1-\hat{p}_2)}{n}\right)$$

$$P_2 - P_1: \hat{p}_2 - \hat{p}_1 \pm \sqrt{\frac{\hat{p}_1(1-\hat{p}_1) + \hat{p}_2(1-\hat{p}_2)}{n}}$$

NPS

0 1 2 3 4 5 6 7 8 9 10

гетракторы

промоутеры

нейтралы

A

N

L

$$X_1, \dots, X_n \sim \text{iid}$$

$$A + N + L = n$$

$$P_A \quad P_N \quad P_L$$

$$NPS = P_L - P_A$$

$$\text{Cov}(P_L, P_A) < 0$$

$$\hat{NPS} = \hat{P}_L - \hat{P}_A$$

NPS - г.ч. - ?

$$\hat{P}_L \overset{\text{asy}}{\sim} \mathcal{N}\left(P_L; \frac{\hat{P}_L(1-\hat{P}_L)}{n}\right)$$

$$\hat{P}_A \overset{\text{asy}}{\sim} \mathcal{N}\left(P_A; \frac{\hat{P}_A(1-\hat{P}_A)}{n}\right)$$

$$\hat{NPS} \overset{\text{asy}}{\sim} \mathcal{N}\left(P_L - P_A; \ast\right)$$

$$\frac{\hat{P}_L(1-\hat{P}_L)}{n} + \frac{\hat{P}_A(1-\hat{P}_A)}{n} - 2 \cdot \left(-\frac{\hat{P}_L \hat{P}_A}{n}\right)$$

$$\text{Var}(\hat{P}_L) + \text{Var}(\hat{P}_A) - 2 \text{Cov}(\hat{P}_L, \hat{P}_A)$$

$$\text{Cov}(\hat{P}_L, \hat{P}_A) = \text{Cov}\left(\frac{L}{n}, \frac{A}{n}\right) = \frac{1}{n^2} \text{Cov}(L, A)$$

$$\underbrace{L + A + N}_{\text{const}} = n$$

$$\text{Cov}(\underbrace{L+A+N}_n, A) = 0$$

$$\underset{C_{LA}}{\text{Cov}(L, A)} + \underset{V_A}{\text{Var}(A)} + \underset{C_{NA}}{\text{Cov}(N, A)} = 0$$

$$\begin{cases} V_A + C_{LA} + C_{NA} = 0 \\ V_L + C_{LA} + C_{LN} = 0 \\ V_N + C_{LN} + C_{NA} = 0 \end{cases} \Leftrightarrow \dots$$

$$C_{LA} = \frac{-V_A + V_N - V_L}{2} = -n P_L P_A$$

Конечная генеральная совокупность

$x_1, \dots, x_N$ — ген. сов. μ, σ^2

$X_1, \dots, X_n$ — выборка БЕЗ повторений

выборка не i.i.d

$$\text{Cov}(X_1, X_2) < 0$$

$$n < N$$

$\bar{X}$

$$\mathbb{E}(\bar{X}) = ? \quad \text{Var}(\bar{X}) = ?$$

~~$$P(X_i = x_s) = P(X_j = x_s) = \frac{(N-1)!}{N!} = \frac{1}{N}$$~~

~~$$E(X_i) = E(X_j)$$~~

$$E(\bar{X}) = \mu$$

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \left(\text{Var}(X_1) + \dots + \text{Var}(X_n) + 2 \text{Cov}(X_1, X_2) + \dots \right) =$$

$$= \frac{1}{n^2} \left(n \cdot \text{Var}(X_i) + n(n-1) \cdot \text{Cov}(X_i, X_j) \right) =$$

$$= \frac{\sigma^2}{n} - \frac{(n-1)}{N-1} \cdot \frac{\sigma^2}{n} = \left(1 - \frac{n-1}{N-1} \right) \frac{\sigma^2}{n}$$

поправка на конечность
рек. сов.

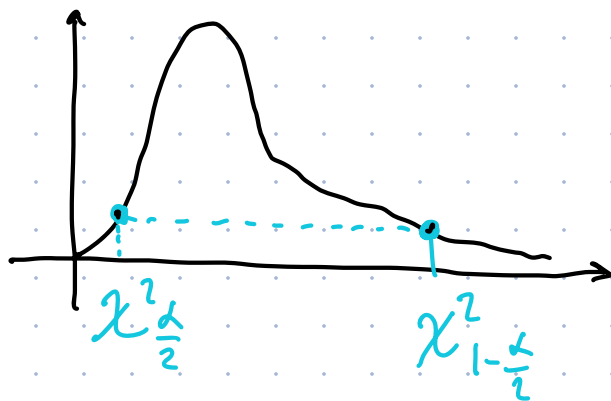
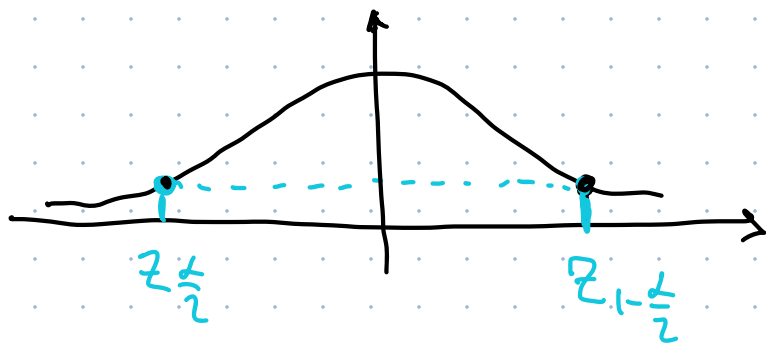
$$\text{Cov}(\overbrace{X_1 + \dots + X_N}^{\text{const}}, X_i) = 0$$

$$\text{Var}(X_1) + (N-1) \cdot \text{Cov}(X_i, X_j) = 0$$

$$\text{Cov}(X_i, X_j) = -\frac{\text{Var}(X_i)}{N-1} < 0 \quad -\frac{\sigma^2}{N-1}$$

Если $\frac{n}{N} \approx 0$ тогда на поправку можно

Самые короткие г.ч.



$f(t)$ - плотность р.-р.

$$\gamma \in (0, 1)$$

$[a; b]$ - интервал мин. длины
из тех, для которых $\int_a^b f(t) dt = \gamma$

γ мин всегда $f(a) = f(b)$

$$\begin{cases} b-a \rightarrow \min_{a,b} \\ \int_a^b f(t) dt = \gamma \end{cases} \quad F(b) - F(a) = \gamma$$

$$\mathcal{L} = (b-a) + \lambda \cdot (F(b) - F(a) - \gamma)$$

$$\frac{\partial \mathcal{L}}{\partial a} = -1 - \lambda \cdot \frac{F'(a)}{f(a)} = 0$$

$$\frac{\partial \mathcal{L}}{\partial b} = 1 + \lambda \cdot \frac{F'(b)}{f(b)} = 0$$

Сложив
уравнения



$$\lambda \cdot (f(b) - f(a)) = 0$$

$$\lambda \neq 0 \Rightarrow f(b) - f(a)$$